FOR A PROPER MORPHISM THE SMALL PUSHFORWARD COMPOSES WITH THE PH-COMPARISON TO THE BIG ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reflection of isomorphisms, the two descriptions of the pullbacks in the ph-setting, and the identification of the composite with part (1) of the comparison lemma.

Editorial note. The label lemma-proper-push-pull-ph-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement refers to the standing comparison of the etale and ph topologies and is delivered verbatim; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Part (1) of the comparison of the small and big higher direct images. Same chapter, the comparison lemma for proper morphisms, giving pi_S^{-1} f_small,* = f_big,* pi_T^{-1}. Supplies the isomorphism quoted at the end of the proof.
- The description of the pullback along the comparison to the ph-topology. Same chapter, the lemma describing the pullback along pi in the ph-setting, giving epsilon_{Y,*} a_Y^{-1} = pi_Y^{-1}. Supplies the identifications of both sides with the corresponding etale statements.

Source reference. Original reference 0DDJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21323–21323. Prerequisite pointer invoked at proof line(s) 22297; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22175–22175. Prerequisite pointer invoked at proof line(s) 22301; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDX: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 22260–22260. Prerequisite pointer invoked at proof

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line(s) 22291; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Lemma 0DDX if f is proper, then we have $a_Y^{-1} \circ f_{small,*} = f_{big,ph,*} \circ a_X^{-1}$.*

Proof. You can prove this by repeating the proof of Lemma 0DDJ part (1); we will instead deduce the result from this. As $\epsilon_{Y,*}$ is the identity functor on underlying presheaves, it reflects isomorphisms. The description in Lemma 0DDW shows that $\epsilon_{Y,*} \circ a_Y^{-1} = \pi_Y^{-1}$ and similarly for X. To show that the canonical map $a_Y^{-1} f_{small,*}\mathcal{F} \to f_{big,ph,*} a_X^{-1}\mathcal{F}$ is an isomorphism, it suffices to show that

$$\begin{aligned} \pi_Y^{-1} f_{small,*}\mathcal{F} &= \epsilon_{Y,*} a_Y^{-1} f_{small,*}\mathcal{F} \\ &\to \epsilon_{Y,*} f_{big,ph,*} a_X^{-1}\mathcal{F} \\ &= f_{big,étale,*}\epsilon_{X,*} a_X^{-1}\mathcal{F} \\ &= f_{big,étale,*}\pi_X^{-1}\mathcal{F} \end{aligned}$$

is an isomorphism. This is part (1) of Lemma 0DDJ. □